HSP&F **Saudi Arabia:** A Health System Analysis

Executive Summary

Saudi Arabia's health system has achieved significant progress, including increased life expectancy, reduced infant mortality, and expanded healthcare infrastructure. Free healthcare for citizens, supported by substantial public spending from oil revenues, has driven near-universal health coverage and strides toward equitable access. Vision 2030 and digital health initiatives aim to enhance efficiency and diversify funding. Challenges include workforce shortages, urban-rural disparities, and reliance on expatriate workers and oil revenues, which pose sustainability risks. Rising healthcare costs from an aging population and chronic diseases further strain the system. Despite reforms, fragmented service delivery and inequitable resource allocation persist. Recommendations focus on increasing Saudization of the workforce, improving resource allocation, enhancing expatriate worker conditions, and integrating mental health education to build a more resilient and equitable healthcare system.

Demographics & Epidemiology

85% of the Saudi population reside in urban centres, resulting in advanced healthcare infrastructure within cities but disparities in rural areas. Expatriates constitute ~38% of the population, ensuring healthcare access for non-citizens (who are often reliant on private insurance) and addressing disparities in coverage compared to citizens.

Saudi's population is currently considered a young population, with 63% of its population <30 years, driving demand for maternal and child health services, preventative care, and future-focused health infrastructure.

However, it is experiencing a demographic shift towards an aging population, driven by factors such as increasing life expectancy and declining fertility rates. The former has almost doubled since the 1960s from ~46 years to ~75 years, and is expected to reach 83 years by 2050. Such a shift will lead to increased rates of chronic diseases, further necessitating a shift from Saudi's current curative-based to preventative-based care.

Socio-Cultural

There are significant differences in the number of primary healthcare centres (PHCs) per 100,000 people across the country's 20 administrative regions. The capital, Riyadh, has a high ration of 7.5 PHCs per 100,000, indicating better access to primary healthcare services. In contrast, Jeddah, has only 3.2 PHCs per 100,000, despite being a populous city. A decline in the overall ratio of PHCs per capita suggests a growing strain on primary healthcare resources as the population increases and is projected to age.

Technological

Saudi's digital health market growth at a CAGR of 21.3% reflects the county's commitment to integrating technology into healthcare delivery, in accordance with Vision 2030 goals. These can enhance patient management by providing real-time monitoring of chronic diseases, improve diagnostic accuracy, and facilitate more effective treatments.

Political

The country has transitioned from a traditional absolute monarchy to a modern authoritarian state. While the royal family maintains centralised control, the kingdom is now actively advancing a more dynamic, efficient, and modern healthcare system.

Legal & Regulatory Environment

The legal system combines Islamic law, royal decrees, and modern governance, with the King holding absolute authority over legislation, regulation, and enforcement. The judiciary operates under royal oversight, and the regulatory framework aligns with the royal family's interests and priorities.

Economic

Saudi Arabia's oil-driven economy, though boasting a high GDP per capita, remains vulnerable to oil price fluctuations. While oil revenues have historically funded free public healthcare, the kingdom is now pursuing economic diversification, decreasing dependence on oil.

Ecological

Extreme heat, poor air quality, water scarcity, and urbanisation contribute to Saudi Arabia's disease burden. Ensuring healthcare resilience against climate change while promoting equitable access to care is key to achieving ecological sustainability.

Performance

	KSA	UAE	Oman	Qatar
Life expectancy at birth (years)				
Male	74.9	76.1	77.2	77.3
Female	77.3	78.3	79.0	79.4
Infant mortality (per 1,000 live births)	5.3	5.6	6.7	9.5
HALE at birth (years)	65.0	66.5	68.0	67.5
HALE at 60 (years)	15.2	16.0	16.5	16.0
Annual Total Health Expenditure (US$/Capita)	1,147	1,200	1,800	1,000
Health Expenditure as Percentage of GDP (%)	5.7	4.0	2.2	4.3
Physician-to-Population ratio (per 100,000 people)	3.077	2.5	3.7	2.0

Goals & Challenges

Goals:

1. Develop health clusters and e-health innovation
2. Attract investments and promote privatisation in the healthcare sector
3. Enhance healthcare services and foster preventative care measures
4. Expand healthcare infrastructure, adding 26,000 to 43,000 hospital beds by 2030.

Challenges:

1. Growing burden of noncommunicable diseases like diabetes and obesity, straining the healthcare system
2. Inequitable access to health services between urban and rural areas
3. shortage of healthcare professionals, especially in specialised fields
4. Need for improve quality and safety of services
5. Gaps in the referral system and coordination between different levels of care.

Financial Design

- The healthcare system is mainly publicly funded.
- Oil industry funds most healthcare spending, and the value-added tax (VAT) introduced in 2018 generates new revenue, though it raises fairness concerns due to its regressive nature.
- The private sector contributes to a smaller yet gradually increasing share through mandatory private insurance, OOPs, and employer contributions.
- A low debt-to-GDP ratio of 26.23% indicates good fiscal sustainability, but the rapid rise in health spending and global fluctuations in oil prices are long-term threats.
- Fixed and demand-isolated pooling of healthcare funds in the public sector; fragmented and risk-based in the private sector, limiting efficiency and equity in coverage.
- "Vision 2030" focuses on diversifying income sources and reducing public spending through privatization and public-private partnerships.



68.92% Government
16.96% Private Health Insurance
14.12% Out Of Pocket (OOP) Spending

Figure 2: Sources of healthcare financing in Saudi Arabia (2023). Adapted from National Health Accounts 2022-2023.

Figure 3: Oil prices (in USD) and the Saudi Arabian government and MOH budgets (in millions of SAR) from 1990 to 2019. Adapted from Alshammari et al., 2023

Provider Payments

Public Sector
- Global Budgeting - MOH healthcare facilities receive a fixed annual budget
- Salaried Payments - reduces financial incentives to overprovide services
- Service-Based Reimbursement - some specialist hospitals receive funding based on the volume and complexity of certain services

Private Sector
- Fee-for-service - reimbursed based on services rendered which can result in increased costs due to overutilization
- Insurance-Based Payments - many hospitals receive payments negotiated with insurers
- Capitation - emerging method some private hospitals are experimenting with

Vision 2030 Payment Reform
- Value-Based Payment - Saudi government shifting towards this model
- Diagnosis-Related Groups (DRGs) - Method being explored to standardise reimbursement and efficiency

Stewardship & Organisation

- Saudi Arabia operates a vertically integrated and segmented, mixed system.
- The Ministry of Health (MoH) is the primary service provider, financer, and regulator, covering citizens, expatriates in the public sector, and pilgrims during Hajj and Umrah.
- Other government entities provide healthcare through separate governance and budgets, competing with the MOH for budget allocations, and creating variations in funding levels and policies.
- Private healthcare insurance is compulsory for all employees working in the private sector under the cooperative health insurance (CHI) programme.
- The MOH has a semi-decentralized structure, with regional directorates having relative autonomy in planning and staffing but limited financial authority.
- Under "Vision 2030", the KSA is moving towards a more decentralized structure through privatisation, corporatization, and digitalization, with the CHI expanding to cover all citizens, residents, and visitors.

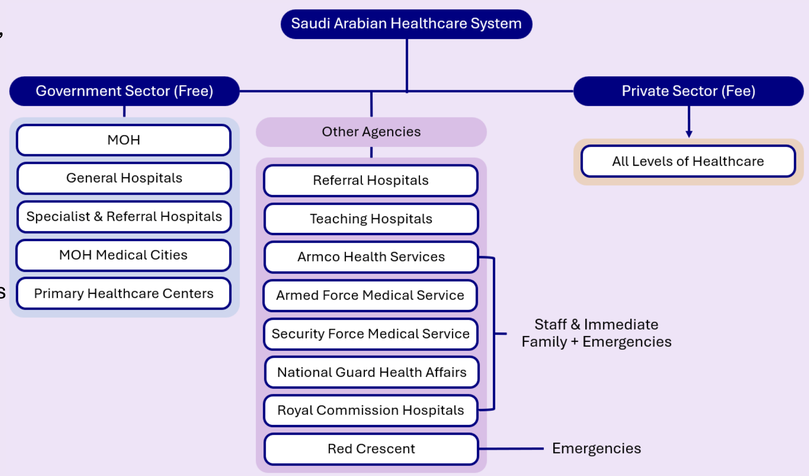


Figure 4: Current structure of the health care sectors in Saudi Arabia, Adapted from Ministry of Health, General Directorate of Hospitals

Service Delivery

Comprehensiveness

The Council of Cooperative Health Insurance(CCHI) determines the services covered by standard health insurance. Primary care, specialist services, hospital care, and prescription medication are included.

Accessibility

In Saudi Arabia, accessibility varies between urban and rural areas. Out-of-pocket health expenditure accounts for 9.93% of current health expenditure, indicating some financial barriers remain. Despite this, nearly all citizens have basic coverage.

Quality & Accreditation

- The MOH has accreditation programs to ensure high-quality care
- Health services and digital transformation are improving efficiency and accessibility.

Person-Centeredness

Person-centredness in healthcare is improving, with user satisfaction with outpatient responsiveness report being 81% positive. However, challenges remain in implementing this approach uniformly across all healthcare settings.

Accountability & Efficiency

- Resource allocation varies between regions , Riyadh has 21,500 beds compared to 8,900 in Eastern regions
- Current bed density of 2.2 per 1,000 residents (below global average of 2.7)
- Case Mix Index of 1.26

Coverage

- The Ministry of Health (MOH) delivers free healthcare to all Saudi citizens and public sector employees.
- Other government bodies (e.g., Ministry of Defence, National Guard) provide healthcare to specific populations.

Coordination

In Saudi Arabia , coordination in health service delivery is challenged by a decentralised system. Responsibility is split between the Ministry of Health and other governmental entities, leading to variation in service delivery across regions.

Resource Allocation

In 2023, the Saudi government allocated approximately SR105 billion ($28.35 billion USD) to healthcare, accounting for 9.4% of its budget. However, a strategic goal of Vision 2030 is to reduce public healthcare expenditure by increasing private sector involvement.

- **Privatisation:** Increasing private sector healthcare contribution from 40% to 65% by 2030, with the privatisation of 290 hospitals and 2,300 primary health centres.
- **Health Clusters:** Decentralising efforts to improve efficiency and accessibility with the formation of regional health clusters which integrate various care levels.
- **Digital Health:** Investment of $1.5 billion USD in healthcare related IT, digital transformation and AI.
- **Focus on Non-Communicable Diseases:** 25-35% of the healthcare budget dedicated to diseases like diabetes, obesity, and cardiovascular diseases.
- **Hospital Bed Expansion**: Identified need for approximately 17,000 additional hospital beds to achieve the international standard of 2.7 beds per 1,000 population, resources are being distributed accordingly

Recommendations

Improve Saudization of local Healthcare Workforce:
- Expand nursing and medicine education through regional and corporate partnerships with the Ministry of Education, utilising local development funds.
- Introduce family-friendly policies, flexible shifts, and childcare support, while deploying allied health professionals to reduce nursing workloads.
- Diversifying incentive strategies (access to public land, increasing salary, bonus for night shifts e.g.)

Improve Resource Allocation:
- The decentralisation of the healthcare system has created greater health inequalities in rural areas.
- They had reduced access to specialist care and fewer hospital beds.
- By increasing telehealth and communications with rural healthcare centres it can help reduce the inequality in care

Improve Expatriate Workers' Conditions and Enhancing Recruitment Standards:
- Standardize recruitment processes with international agencies to hire highly skilled workers, avoiding tokenism.
- Provide cultural and language training to support expatriate workers' integration.
- Implement corporate policies promoting a positive working environment and discourage violence, particularly towards expatriate workers.

Integrate Mental Health School and Social Awareness Education:
- Launch social awareness campaigns via social media and podcasts to combat mental health stigma and encourage seeking treatment.
- Integrate mental health education into school curricula, emphasizing it as a treatable condition, not a stigma, and deploy school-based mental health nurses for early support.